DATA ITEM DESCRIPTION
Title: MAINTAINABILITY AND BUILT IN TEST DEMONSTRATION PROCEDURE

Number: DI-SESS-81604B
AMSC Number: N10152
DTIC Applicable: Yes
Defense Technical Information Center
ATTN: DTIC-
8725 John J. Kingman Road, Suite 0944
Fort Belvoir VA 22060-6218
Preparing Activity: AS
Applicable Forms: N/A

Approval Date: 20200218
Limitation: N/A
GIDEP Applicable: Yes
GIDEP Operations Center
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Project Number: SESS-2020-018

Use/relationship: The Maintainability and Built In Test (BIT) Demonstration Procedure provides detailed technical directions for implementing a maintainability or BIT demonstration at the equipment, subsystem, or system levels during the integrated test program. The procedure contains step-by-step instructions for how the equipment, including software, involved is utilized during the test or demonstration. This document is used by the procuring activity for review and approval of the contractor's procedure for conducting the maintainability or BIT demonstration.

This Data Item Description (DID) contains the format, intended use information for the data deliverable resulting from the work task described in the contract.

This DID supersedes DI-SESS-81604A

Requirements:

1. Reference documents. None.

2. Format. The Maintainability and BIT Demonstration Procedure shall be in the contractor's format.

3. Content. The Maintainability and BIT Demonstration Procedure shall contain:

3.1. Test objectives and quantitative goals or requirements to be demonstrated, including specific references to the applicable sections of the specification.

3.2. Description of the test team roles and responsibilities.

3.3. A list and brief description of all units comprising the equipment and a specific listing of those units that are placed on test and the up-to-date configuration (drawing list, including approved changes, waivers , and deviations), including software configuration.

3.4. A list and description of all test and monitoring equipment and software, including: manufacturer, model number, calibration records (where applicable), software configuration information, and facility descriptions and requirements.

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3.5. Allowable adjustments and normal checkout procedures for the equipment under test.

3.6. A description of all preventive maintenance measures to be performed during the test, if such maintenance is permitted by the detailed equipment specification.

3.7. Interconnecting cable diagrams and interface conditions of complete test, setup including equipment under test and test monitoring equipment.

3.8. Performance parameters to be measured, frequency of measurement, and the method used.

3.9. A description of test methods and mode(s) of operation of the items under test, including configuration and mission requirements.

3.10. Procedure for selection of maintenance tasks when faults are simulated.

3.11. Identification of special maintenance tasks.

3.12. The statistical maintainability or BIT design criteria, as appropriate, including decision criteria (fail and accept criteria), if applicable.

3.13. Test duration and conditions, including the duty cycle and environmental, operational, and performance profiles during performance parameter measurement.

3.14. A list of data elements to be recorded during tests and acquisition methods.

3.15. Include preliminary applicable data collection sheets and scoring and evaluation criteria (i.e., Failure Definition and Scoring Criteria), including: the data analysis methods, analytical procedures, and evaluation methodology that are used to compute maintainability or BIT performance based on the data collected during testing.

3.16. A list of the maintenance tasks to be verified, demonstrated, and evaluated.

3.17. Description of anticipated system faults and mapping of fault detection by BIT.

3.18. Description of BIT fault insertion technique(s), including how and where faults are inserted.

3.19. A description of maintenance concept and level(s) of maintenance.

3.20. Description of and assessment of the adequacy of the item support elements, and an estimate of their effect on the item maintainability to include:

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3.20.1. Maintenance planning.

3.20.2. Support equipment, tools, and test equipment.

3.20.3. Supply support, including both spares and consumables.

3.20.4. Transportation, handling, and storage.

3.20.5. Technical data, including technical manuals to be used.

3.20.6. Any applicable safety and calibration equipment.

3.21. The Government furnished property requirements and identification of the impact of use of Government furnished property.

3.22. Identification of the failure Reporting, Analysis and Corrective Action (FRACAS) system to be used.

3.23. Detailed reason for each proposed deviation.

3.24. The test conduct ground rules and identification of the system interface boundaries.

3.25. A description of the action(s) to be taken if a reject decision is reached, including corrective action plan and retest provisions.

End of DI-SESS-81604B

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